

# Addition and subtraction of algebraic fractions

Same denominator — add the numerators. Different denominators — build them the same first, then add.

LESSON 12–14

3 × 40 MIN

ALGEBRA 8, §4

STAGE 9 · 2.5

## LESSON CLOCK

|    |     |    |     |    |
|----|-----|----|-----|----|
| 5' | 12' | 8' | 13' | 2' |
|----|-----|----|-----|----|

Warm-up 5 min

Explanation 12 min

Interactive 8 min

Practice 13 min

Homework 2 min

## LEARNING OBJECTIVES

- Add and subtract fractions with the same denominator.
- Add and subtract fractions with different denominators using the LCD.
- Handle subtraction correctly — bracket the whole numerator being subtracted.
- Simplify the result and state the permissible values.

## TEXTBOOK

National \$4, pp. 22–26

Cambridge Learner's Book pp. 40–44

Workbook Workbook 2.5

## 01 Explanation

### Same denominator

$$\frac{A}{C} \pm \frac{B}{C} = \frac{A \pm B}{C}$$

Keep the denominator, combine the numerators, then **always try to cancel the result** — the answer is often much simpler than the working suggests.

$$\frac{x}{x+3} + \frac{3}{x+3} = \frac{x+3}{x+3} = 1$$

### Different denominators

## FOUR STEPS, EVERY TIME

1. Factorise the denominators and find the LCD.
2. Find each additional factor and build both fractions up.

3. Combine the numerators over the single denominator.
4. Simplify the numerator, cancel if you can, state the permissible values.

$$\frac{1}{x-1} + \frac{1}{x+1} = \frac{(x+1) + (x-1)}{(x-1)(x+1)} = \frac{2x}{x^2-1}$$

### The bracket that everyone forgets

#### SUBTRACTION

When you subtract a fraction, the minus sign belongs to the **whole** numerator, so put it in a bracket before you open it.

$$\frac{2x}{x^2-4} - \frac{1}{x-2} = \frac{2x - (x+2)}{(x-2)(x+2)} = \frac{x-2}{(x-2)(x+2)} = \frac{1}{x+2}$$

Writing  $2x - x + 2$  instead of  $2x - (x + 2)$  changes the answer completely.

### A whole expression is a fraction too

Any expression can be written over the denominator 1, so it can join the sum:

$$1 - \frac{1}{x+1} = \frac{(x+1) - 1}{x+1} = \frac{x}{x+1}$$

## 02 Worked examples

**EXAMPLE 1** *Simplify*  $\frac{5}{x^2 - 9} + \frac{1}{x + 3}$

①  $x^2 - 9 = (x - 3)(x + 3)$

Factorise.

②  $LCD = (x - 3)(x + 3)$

The second denominator is one of its factors.

③ 
$$\frac{5}{(x - 3)(x + 3)} + \frac{x - 3}{(x - 3)(x + 3)}$$

Additional factor  $(x - 3)$  for the second fraction.

④ 
$$\frac{5 + x - 3}{(x - 3)(x + 3)} = \frac{x + 2}{x^2 - 9}$$

Combine and tidy the numerator.

**ANSWER**

$$\frac{x + 2}{x^2 - 9}, \quad x \neq \pm 3$$

**EXAMPLE 2****Simplify**

$$\frac{1}{x-1} - \frac{2}{x^2-1}$$

①  $x^2 - 1 = (x - 1)(x + 1)$

Factorise.

②  $LCD = (x - 1)(x + 1)$

The first denominator needs the factor  $(x + 1)$ .

③ 
$$\frac{x+1}{(x-1)(x+1)} - \frac{2}{(x-1)(x+1)}$$

Build the first fraction up.

④ 
$$\frac{(x+1)-2}{(x-1)(x+1)} = \frac{x-1}{(x-1)(x+1)}$$

Combine — keep the bracket.

⑤ 
$$\frac{1}{x+1}$$

Cancel  $(x - 1)$ . Always look for this last step.**ANSWER**

$$\frac{1}{x+1}, \quad x \neq \pm 1$$

**EXAMPLE 3*****Simplify***

$$\frac{1}{x(x+1)} + \frac{1}{(x+1)(x+2)}$$

①  $LCD = x(x+1)(x+2)$

$(x+1)$  is shared, so it is taken once.

② 
$$\frac{x+2}{x(x+1)(x+2)} + \frac{x}{x(x+1)(x+2)}$$

Additional factors  $(x+2)$  and  $x$ .

③ 
$$\frac{2x+2}{x(x+1)(x+2)}$$

Combine the numerators.

④ 
$$\frac{2(x+1)}{x(x+1)(x+2)} = \frac{2}{x(x+2)}$$

Factorise the numerator and cancel  $(x+1)$ .

**ANSWER**

$$\frac{2}{x(x+2)}, \quad x \neq 0, -1, -2$$

### 03 Interactive model

Reveal the four steps one at a time and ask which step the class thinks comes next.

## Adding and subtracting, step by step

$$\frac{1}{x-1} + \frac{1}{x+1}$$

①  $LCD = (x-1)(x+1)$

Different  
factors,  
so  
multiply  
them.

② *additional factors:  $(x+1)$  and  $(x-1)$*

LCD  
÷  
each  
denominator.

③ 
$$\frac{(x+1) + (x-1)}{(x-1)(x+1)}$$

Build  
both  
up  
and  
combine.

④ 
$$\frac{2x}{x^2-1}$$

Tidy  
the  
numerator.  
Nothing  
cancels  
here.

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH             | ЎЗБЕКЧА                  | РУССКИЙ                           |
|---------------------|--------------------------|-----------------------------------|
| Add                 | Qo'shish                 | Сложение                          |
| Subtract            | Ayirish                  | Вычитание                         |
| Like fractions      | Bir xil maxrajli kasrlar | Дроби с одинаковыми знаменателями |
| Unlike fractions    | Har xil maxrajli kasrlar | Дроби с разными знаменателями     |
| Combine             | Birlashtirish            | Объединить                        |
| Numerator           | Surat                    | Числитель                         |
| Whole expression    | Butun ifoda              | Целое выражение                   |
| Simplify the result | Natijani soddalashtirish | Упростить результат               |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

**Quick check**
**Check yourself**

3 x + 5 x equals:

$$\frac{8}{2x}$$

$$\frac{8}{x}$$

$$\frac{15}{x}$$

$$\frac{8}{x^2}$$

2x x<sup>2</sup> - 4 - 1 x - 2 equals:

$$\frac{1}{x+2}$$

$$\frac{2x-1}{x^2-4}$$

$$\frac{x+2}{x^2-4}$$

$$\frac{3x+2}{x^2-4}$$



1  $a - b$  + 1  $b - a$  equals:

$$\frac{2}{a - b}$$

$$0$$

$$\frac{2}{b - a}$$

it cannot be combined

x  $x + 3$  + 3  $x + 3$  equals:

$$\frac{x + 3}{2x + 6}$$

$$1$$

$$\frac{3x}{x + 3}$$

$$x$$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.  $\frac{3}{x} + \frac{5}{x}$

ANSWER  $\frac{8}{x}$

2.  $\frac{7}{a} - \frac{2}{a}$

ANSWER  $\frac{5}{a}$

3.  $\frac{x}{4} + \frac{x}{4}$

ANSWER  $\frac{x}{2}$

4.  $\frac{2a}{b} + \frac{3a}{b}$

ANSWER  $\frac{5a}{b}$

5.  $\frac{1}{2} + \frac{1}{x}$

ANSWER  $\frac{x+2}{2x}$

6.  $\frac{1}{x} - \frac{1}{y}$

ANSWER  $\frac{y-x}{xy}$

7. 
$$\frac{3}{x+1} + \frac{2}{x+1}$$

ANSWER 
$$\frac{5}{x+1}$$

**Medium · the standard question**

7 PROBLEMS

$$1. \quad \frac{1}{x-1} + \frac{1}{x+1}$$

ANSWER  $\frac{2x}{x^2-1}$

$$2. \quad \frac{1}{x-2} - \frac{1}{x+2}$$

ANSWER  $\frac{4}{x^2-4}$

$$3. \quad \frac{x}{x+3} + \frac{3}{x+3}$$

ANSWER 1,  $x \neq -3$

$$4. \quad \frac{2}{x} + \frac{3}{x^2}$$

ANSWER  $\frac{2x+3}{x^2}$

$$5. \quad \frac{a}{a-b} - \frac{b}{a-b}$$

ANSWER 1,  $a \neq b$

$$6. \quad \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

ANSWER  $\frac{bc+ac+ab}{abc}$

$$7. \quad \frac{5}{x^2-9} + \frac{1}{x+3}$$

ANSWER  $\frac{x+2}{x^2-9}$

## Hard · stretch and reason

7 PROBLEMS

$$1. \quad \frac{1}{x^2 - x} + \frac{1}{x^2 + x}$$

ANSWER  $\frac{2}{x^2 - 1}$

$$2. \quad \frac{1}{x - 3} + \frac{1}{x + 3}$$

ANSWER  $\frac{2x}{x^2 - 9}$

$$3. \quad \frac{1}{a - b} + \frac{1}{b - a}$$

ANSWER  $0, a \neq b$

$$4. \quad \frac{2x}{x^2 - 4} - \frac{1}{x - 2}$$

ANSWER  $\frac{1}{x + 2}$

$$5. \quad 1 - \frac{1}{x + 1}$$

ANSWER  $\frac{x}{x + 1}$

$$6. \quad \frac{1}{x - 1} - \frac{2}{x^2 - 1}$$

ANSWER  $\frac{1}{x + 1}$

$$7. \quad \frac{1}{x(x + 1)} + \frac{1}{(x + 1)(x + 2)}$$

ANSWER  $\frac{2}{x(x + 2)}$

**07 Homework****Homework — 6 problems**

Algebra 8, §4, pp. 22–26. Cancel the answer wherever you can.

1.  $\frac{4}{x} + \frac{9}{x}$

2.  $\frac{1}{x+2} + \frac{1}{x-2}$

3.  $\frac{3}{x-4} - \frac{3}{x+4}$

4.  $\frac{x}{x-5} - \frac{5}{x-5}$

5.  $\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}$

6.  $2 - \frac{3}{x-1}$